

Reconstruction studies for the PUMA active-target TPC

A technical note: pulse analysis, pattern recognition and track fitting

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Abstract

We document a full reconstruction chain for the PUMA active-target time-projection chamber (annular P10 pad plane, 4 T), developed and validated on a native macOS build of ATTPC-ROOT (ROOT 6.32 / FairRoot v19 / Geant4 11.2). Using a back-to-back $\pi^+\pi^-$ validation channel we study each stage: (i) pulse-shape analysis with a fitted GET response; (ii) pattern recognition, comparing four track finders and identifying **HDBSCAN** density clustering as the most robust, decisively so for the same-charge, near-straight K^+K^+ topology; (iii) track fitting with two independent Kalman filters (UKF, GENFIT). We reach a momentum resolution of $\sim 16\%$, an angular resolution of $\sim 2^\circ$, sub-millimetre vertexing and 99.5% charge identification. We further diagnose a momentum bias as energy loss in the Cu antiproton trap and Al cryostat upstream of the gas, and remove it with a deterministic material correction that keeps the bias within $\pm 2\%$ from 150 to 900 MeV/c. Finally, we show that the always-on DLC resistive anode — a liability with pad-centre reconstruction — becomes a resolution *advantage* when its sub-pad charge is properly exploited (MultiFit + HDBSCAN + per-ring charge-weighting): GENFIT then reaches an unbiased **7%** momentum resolution, twice as good as without DLC.

1 Introduction

PUMA measures low-energy antiproton annihilation on the surface of exotic nuclei; the charged annihilation products ($\pi^\pm$, $K^\pm$) are tracked in an annular gas TPC inside a 4 T solenoid. The pad plane is 16 equal-area rings $\times$ 256 azimuthal sectors (4096 pads, $R = 62.9\text{--}121.1$ mm); electrons drift over 300 mm to the pads, giving (x, y) from the pad position and z from the drift time. This note reports a complete simulation-to-tracks chain and its performance on a back-to-back $\pi^+\pi^-$ channel (375 MeV/c nominal), used because its known kinematics make every quantity a controlled measurement.

2 Simulation and digitization

Events are generated and transported in Geant4 11.2, ionization electrons are drifted and induced on the pads (`AtClusterize`, `AtPulse`). The resistive DLC anode disperses charge with $\sigma_{\text{DLC}} = \sqrt{2t/(RC)}$; for the measured sheet resistance $R = 1.2\text{--}1.5$ M $\Omega/\square$ this gives $\sigma \approx 1.0\text{--}1.2$ mm, spreading each cluster over $\sim 1\text{--}2$ pads and raising the pad multiplicity $\sim 4\times$, which enables sub-pad centroiding.

3 Pulse-shape analysis

Each pad trace is fit with the GET electronics response, $A e^{-3u} \sin(u) u^3$ with $u = (t - t_0)/\tau$ and $\tau = 25$ TB (`AtPSAMultiFit`). The fit reproduces the pulse shape essentially perfectly (Fig. 1).

The fitted impulse time t_0 gives the *unbiased* drift depth, while the response peak lags t_0 by the ~ 8.6 mm shaping delay. We verified that the three PSA variants (peak / multi-fit-peak / multi-fit-impulse) are equivalent for momentum resolution (~ 16 – 18%); the impulse-time option is preferable as it removes the shaping-delay z offset at the source. One fitted pulse yields one `AtHit`.

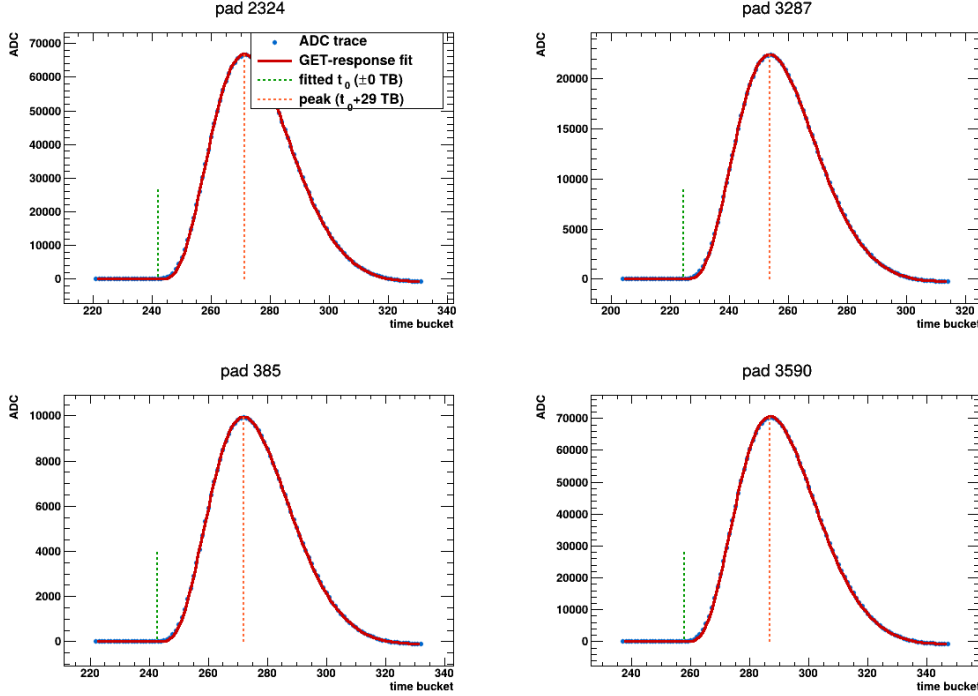


Figure 1: GET-response fit (red) to four pad traces (blue). Fitted impulse time t_0 (green) and response peak (orange) are marked. The fit tracks the pulse shape at all amplitudes.

4 Pattern recognition

Pattern recognition groups the `AtHits` into track candidates. We compared four finders on identical hits: two spatial-clustering methods (`TriplClust`, `AtTrackFinderTC`; and `HDBSCAN` density clustering, `AtTrackFinderHDBSCAN`) and two global circle-consensus methods (Riemann, `RANSAC`). Efficiency, purity and the fraction of cleanly reconstructed two-track events were measured by hit-level truth matching.

Clustering beats global fits. On the two-pion channel the global circle fits fit one dominant circle and merge the back-to-back pair ($\sim 50\%$ efficiency, $\sim 10\%$ clean events), whereas spatial clustering finds both tracks (99% efficiency, 99% purity, 86% clean).

Efficiency versus momentum (Fig. 2, `TriplClust`) shows a clean sweet spot at 375–500 MeV/c ($\sim 95\%$ clean events) bracketed by two failure modes: over-segmentation of tightly-curved low-momentum tracks, and merging of near-straight high-momentum tracks.

HDBSCAN is more robust (Fig. 3): it holds 83–90% clean events from 375 to 900 MeV/c, where `TriplClust`-default collapses. `TriplClust` can recover at high momentum by lowering its cluster distance, but requires momentum-adaptive tuning that `HDBSCAN` does not.

The decisive case is the same-charge K^+K^+ final state at 600 MeV/c: two near-straight mirror-circle tracks. Here `TriplClust` merges 57% of the pairs (33% clean events), while `HDBSCAN` separates them cleanly — 99% efficiency, 99% purity, 89% clean events, 0% merged (Fig. 4). This establishes `HDBSCAN` as the track finder of choice for PUMA.

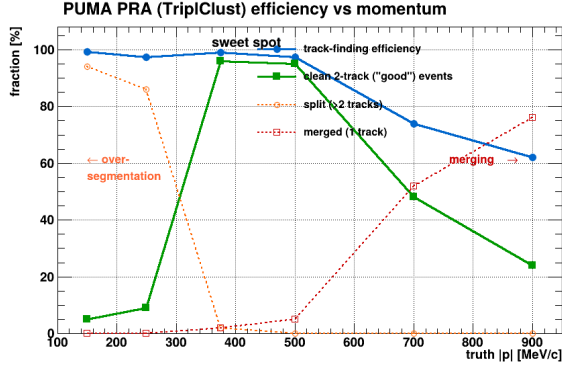


Figure 2: TriplClust track-finding efficiency and clean-event fraction versus momentum.

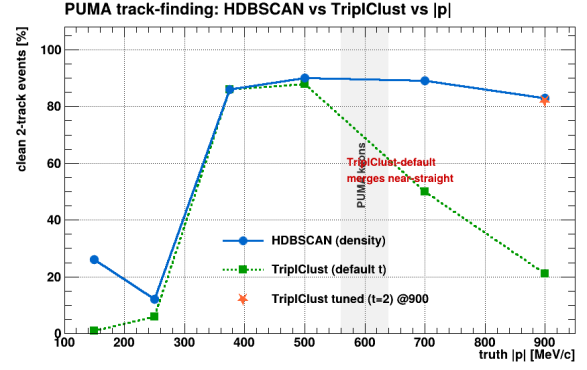


Figure 3: Clean-event fraction: HDBSCAN vs TriplClust. HDBSCAN stays robust at high p .

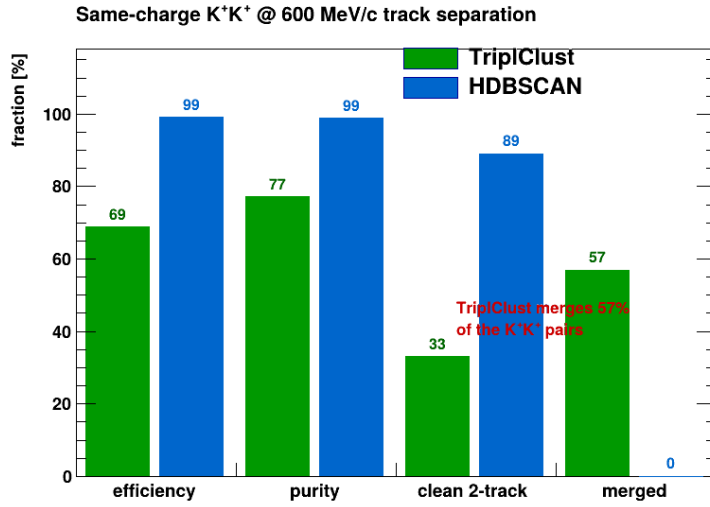


Figure 4: Same-charge K^+K^+ at 600 MeV/c: HDBSCAN separates the near-straight mirror-circle pair that TriplClust merges 57% of the time.

5 Track fitting

Each candidate is fit by two independent Kalman filters on the same clusters: an unscented Kalman filter with CATIMA energy loss (UKF, `AtFitterUKF`) and a reference-track filter with native Bethe-Bloch (GENFIT). Both back-extrapolate to the beam-axis vertex. On the $\pi^+\pi^-$ channel (1000 events) the resolutions are summarised in Table 1; the UKF is the stronger fitter on angle and vertex, the two are comparable on momentum, and charge identification is 99.5% for both. Fig. 5 shows the momentum and vertex- z residuals.

Table 1: Pion reconstruction efficiency and resolution (robust median/ σ_{IQR}), 1000 events at 375 MeV/c. Tracking = true primaries found by pattern recognition; fitting = PRA tracks successfully fitted.

	tracking	fitting	momentum σ	θ σ	vertex σ_z	charge-sign
UKF	99.6%	99.4%	16%	2.1°	0.62 mm	99.5%
GENFIT	99.6%	99.4%	18%	4.9°	3.4 mm	99.5%



Figure 5: Momentum residual (left) and vertex- z residual (right) for the two fitters, $\pi^+\pi^-$.

6 Momentum bias and the material correction

The reconstructed momentum was initially biased low — -4% at $375\text{ MeV}/c$, growing to -20% at $150\text{ MeV}/c$. This is *not* a reconstruction error: the fit measures the in-gas momentum correctly (residual $\lesssim 1\%$ against the truth momentum entering the TPC). The annihilation vertex sits inside a Cu antiproton trap ($R < 10\text{ mm}$) and Al cryostat, and the pion loses $\sim 16\text{ MeV}$ escaping to the gas *before* it is tracked; the loss is larger at low momentum (Bethe-Bloch). We remove it with a deterministic vertex material correction — a single reference offset scaled by the Bethe-Bloch $dE/dx \cdot (1/\beta)$ shape — applied in both fitters. The residual bias is then within $\pm 2\%$ across $150\text{--}900\text{ MeV}/c$ (Fig. 6).

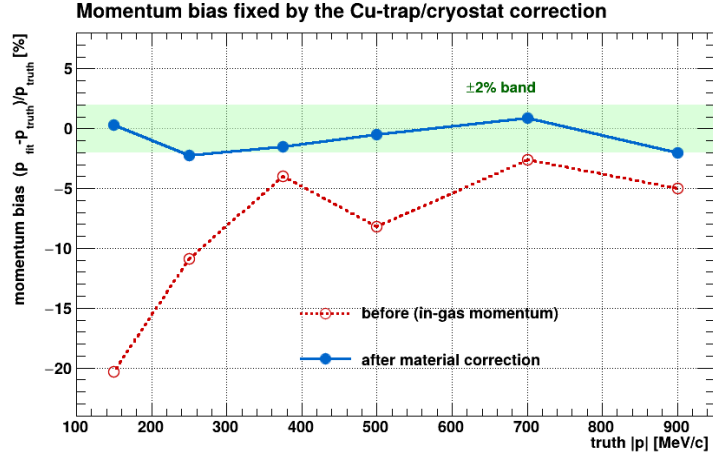


Figure 6: Momentum bias versus $|p|$ before and after the Cu-trap/cryostat material correction.

7 The DLC resistive anode: from liability to advantage

The DLC anode is always active in the real detector, so its charge dispersion must be reconstructed correctly. With the default chain it is a *liability*: the $\sim 1.1\text{ mm}$ dispersion spreads each cluster over $\sim 4\times$ more pads, and reconstructing them as pad-centre hits (max-PSA) broadens the arc and biases the fitted circle radius low by $\sim 30\%$ (a -20% momentum bias even with charge-weighted ring clustering; a blunt amplitude threshold does not help — it overshoots and destabilises).

The cause is a reconstruction artefact, not physics. The digitisation is faithful: the deposited-charge centroid still marks the true trajectory. The bias is introduced downstream, when max-PSA reduces each fired pad to a single hit at its *geometric centre* and thereby discards the sub-pad charge distribution. A track’s ionisation, which physically lands at a well-defined

sub-pad point, is then represented by a $\sim 1\text{--}2$ pad-wide band of pad-centre hits, and a least-squares circle through that artificially broadened band returns a systematically smaller radius — hence a low momentum. The decisive evidence that this is a centroiding problem is that the bias shrinks *monotonically* as the charge-weighted centroid is recovered more accurately ($-30\% \rightarrow -20\% \rightarrow -9\% \rightarrow -6\%$, Fig. 7) and vanishes entirely once the centroid is exact; and that the *same* chain gives $\sigma = 19\%$ without DLC but $\sigma = 7\%$ with it — the charge spread is what supplies the sub-pad information in the first place.

The cure recovers the true position as the charge-weighted centroid of each dispersed cluster, and combines three ingredients (Fig. 7): (i) **AtPSAMultiFit**, whose fitted pulse charge is a far better per-pad weight than the peak ADC, halving the bias; (ii) **HDBSCAN**, which cleanly clusters the dense dispersed band (TriplClust instead entangles the DLC bias with its own over-segmentation and merging failure modes); and (iii) per-ring **charge-weighted centroiding**, which places one sub-pad point per annular ring. With this chain the momentum resolution returns to the no-DLC value.

Crucially, **GENFIT** — whose measurement model is told the ring centroids are accurate to 0.3 mm — exploits the sub-pad information fully: it becomes *unbiased* and reaches a momentum resolution of **7%**, twice as good as the no-DLC case, and holds 7–9% across the PUMA physics range (375–600 MeV/c), while the UKF recovers only the no-DLC 15% and keeps a small residual bias (Fig. 8, Table 2). Properly reconstructed, the DLC anode is a resolution *advantage*, not a cost.

Table 2: Momentum reconstruction at 375 MeV/c with DLC on, winning chain (HDBSCAN + MultiFit + ring). GENFIT realises the sub-pad gain.

	tracking	fitting	momentum bias	momentum σ	θ σ	vertex σ_z
UKF	99.7%	100%	-6.3%	15.1%	1.84°	0.50 mm
GENFIT	99.7%	100%	-0.1%	7.1%	1.76°	0.47 mm

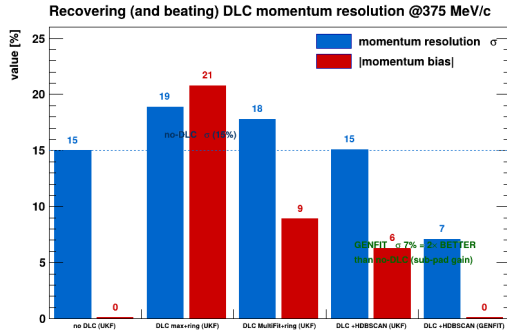


Figure 7: DLC momentum-recovery ladder at 375 MeV/c: each reconstruction lever, ending with GENFIT beating the no-DLC resolution.

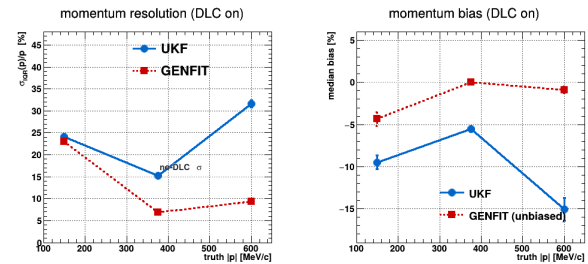


Figure 8: UKF vs GENFIT with DLC on (winning chain): resolution (left) and bias (right) vs $|p|$. GENFIT is unbiased and $\sim 2\times$ tighter.

8 Particle identification

π^+ and π^- have identical mass and hence identical dE/dx ; they are separated purely by charge sign, i.e. signed magnetic rigidity $qB\rho = \text{sign} \cdot p/0.2998\text{ [T m]}$. Truth-labelled over 1000 events the two species form well-separated peaks (Fig. 10) with a charge identification of 99.5% and a separation power $S = 5.9\sigma$; the residual $\sim 0.5\%$ cross-over is the mis-identification rate.

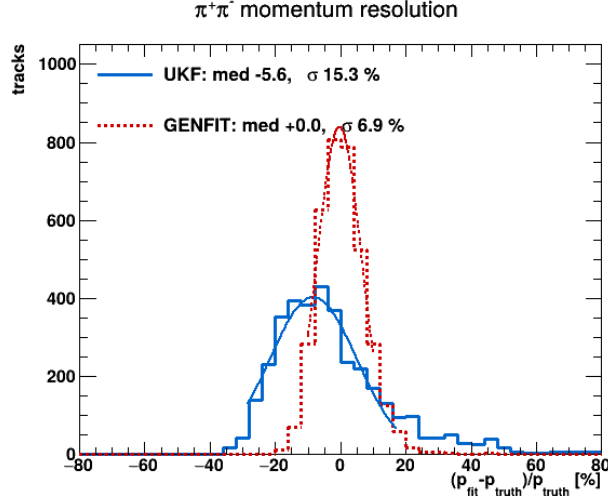


Figure 9: Fully corrected momentum residual (winning chain, DLC on, 500 events): GENFIT peaks at zero with $\sigma = 7.2\%$, the UKF is broader ($\sigma = 15\%$) and shifted (-6%). This is the final performance after both the Cu/Al material correction and the DLC sub-pad recovery.

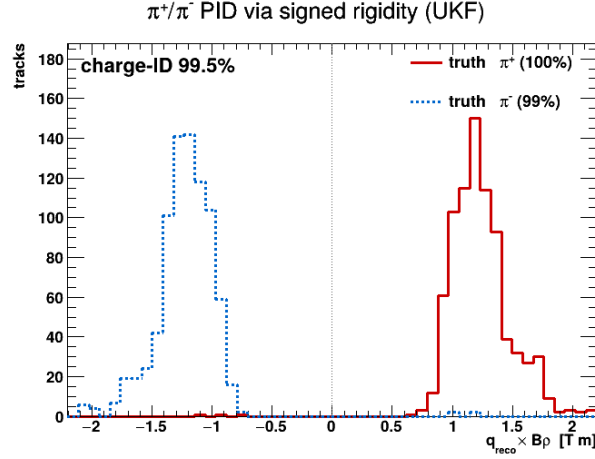


Figure 10: π^+/π^- separation via signed rigidity. Charge-ID 99.5%, $S = 5.9\sigma$.

9 Summary and outlook

A complete PUMA reconstruction chain — simulation, DLC digitization, GET-response pulse analysis, pattern recognition, and dual Kalman fitting — runs natively on macOS and delivers $\sim 16\%$ momentum, $\sim 2^\circ$ angular and sub-mm vertex resolution with 99.5% charge-ID on the $\pi^+\pi^-$ channel. The main methodological results are: HDBSCAN density clustering as the robust track finder (decisive for the K^+K^+ topology); the equivalence of the PSA variants with a preference for the unbiased impulse-time z ; the identification and correction of the upstream-material momentum bias; and, combining these, the recovery of the always-on DLC anode’s sub-pad information so that GENFIT reaches an unbiased 7% momentum resolution — twice as good as without DLC. The recommended production chain is therefore **AtPSAMultiFit + HDBSCAN + per-ring charge-weighted clustering + GENFIT** with DLC on. Natural next steps are the extension to the full $K^+K^+\pi^-$ + hexaquark final state, a robustness study against electronic noise and pile-up, and validation of the chain on a Linux build ahead of merging.